Electric Vehicle Market Performance & Insights

This project analyses 100,000 electric vehicle records (772.6 million annual units sold, 200 manufacturers, 15 categories, 10 manufacturing countries) in Tableau and answers the three stakeholder scenarios with one interactive dashboard and a data story. Screenshot placeholders are marked [Insert screenshot] for you to fill from Tableau.

1. Introduction

The electric vehicle (EV) industry is one of the fastest-growing sectors of the global automotive market, driven by environmental awareness, government incentives, battery advances and demand for sustainable transport. Manufacturers now compete across hatchbacks, sedans, SUVs, buses, trucks and two- and three-wheelers, using battery chemistries such as LFP, NMC, NCA, Sodium-Ion and Solid-State.

The industry produces large volumes of data on sales, pricing, battery specifications, manufacturing locations and customer ratings. Turning that data into dashboards lets manufacturers, consultants and policymakers track annual sales, battery capacity, driving range, customer satisfaction, manufacturing distribution and market share.

2. Problem Statement

EV decision-makers have plenty of data but no single view of which categories sell most, which manufacturers lead, which battery technologies dominate, where vehicles are built and how satisfied customers are. This project builds an interactive Tableau dashboard and story that answers those questions for three personas.

3. Objectives

- Identify which vehicle categories contribute the most to annual sales.
- Measure manufacturer market share and customer satisfaction.
- Compare battery technologies on adoption, lifespan, degradation and range.
- Compare driving range across vehicle categories.
- Map manufacturing distribution across countries.
- Find high-growth and declining segments over model years 2018 to 2026.

- Publish the dashboard to Tableau Public and embed it in a Flask web page.

4. Stakeholder Scenarios

Scenario	Persona	What they need from the dashboard
1	Emily Johnson, EV Market Strategy Director	Category contribution to sales, leading manufacturers, satisfaction by segment, battery adoption, range by category, high-growth segments
2	David Chen, Automotive BI Manager	Manufacturer market share, annual sales, battery distribution, manufacturing-country analysis, customer ratings
3	Dr. Sarah Williams, Sustainable Mobility Research Analyst	Battery technology comparison, range improvement, global manufacturing distribution, satisfaction by category

5. Dataset Description

The file ev_market_test.csv has 100,000 rows and 37 columns, with no duplicate vehicle IDs and no duplicate rows. Key fields are grouped below.

Group	Columns
Identity	vehicle_id, model_name, manufacturer, model_year, manufacturing_country, market_region, category
Battery	battery_capacity_kwh, battery_type, battery_cycle_life, battery_degradation_pct_5yr
Performance	range_km, efficiency_wh_per_km, motor_power_kw, torque_nm, num_motors, drivetrain, top_speed_kmph, acceleration_0_100_sec

Group	Columns
Charging	max_dc_charging_kw, charging_dc_fast_time_min, charging_ac_time_hr, num_charging_ports
Design and cost	num_seats, weight_kg, cargo_capacity_liters, price_usd, insurance_cost_usd_yearly, warranty_years, resale_value_pct_3yr
Quality and sustainability	safety_rating_stars, autonomous_level, infotainment_screen_inch, co2_saved_tons_per_year, customer_rating
Sales and use	annual_sales_units, recommended_use_case

Headline numbers computed from the file

KPI	Value
Vehicle records	100,000
Unique model names	17,919
Manufacturers	200
Total annual sales	772,608,124 units
Average customer rating	4.08 / 5
Average driving range	324.31 km
Average battery capacity	78.23 kWh
Average price	$65,010

Note: the course brief quotes 200,000 rows, 39 columns and 17,980 models. The file supplied is a 100,000-row, 37-column test extract, so every figure in this report comes from that file.

6. Data Connection and Preparation

1. Open Tableau, choose Connect > Text file and select ev_market_test.csv.

2. Confirm data types: model_year as Number (whole), all metric columns as Number (decimal), all names and categories as String. Right-click model_year and set Convert to Dimension so it behaves as a year axis.
3. Use a live or extract connection; an extract (.hyper) keeps 100,000 rows fast.
4. Check for duplicates and missing values (results below).
5. Create the calculated fields in section 7.

Data quality check

Check	Result	Action
Duplicate vehicle_id	0	None
Fully duplicated rows	0	None
Missing annual_sales_units	989 (1.0%)	Left as Null; SUM and AVG ignore nulls
Missing customer_rating	1,931 (1.9%)	Left as Null; AVG ignores nulls
Missing battery_cycle_life	2,028 (2.0%)	Left as Null
Missing resale_value_pct_3yr	3,057 (3.1%)	Not used in dashboard
Missing infotainment_screen_inch	1,506 (1.5%)	Not used in dashboard
Outliers (sales up to 772,442 units, buses with 65 seats)	Genuine large-vehicle and best-seller values	Kept; use median where skew matters

The nulls are under 4% in every column and sit outside the main KPI logic, so no rows were removed.

7. Calculated Fields and Parameters

Create these in Tableau (Analysis > Create Calculated Field).

Market Share %

```
SUM([annual_sales_units]) / TOTAL(SUM([annual_sales_units]))
```

Rating Band (for the rating heatmap)

```
IF [customer_rating] >= 4.5 THEN "Excellent (4.5-5.0)"
ELSEIF [customer_rating] >= 4.0 THEN "Good (4.0-4.49)"
ELSEIF [customer_rating] >= 3.0 THEN "Average (3.0-3.99)"
ELSE "Poor (below 3.0)"
END
```

Sales Growth % vs First Year (table calculation, addressing Model Year)

```
(SUM([annual_sales_units]) - LOOKUP(SUM([annual_sales_units]), FIRST()))
/ ABS(LOOKUP(SUM([annual_sales_units]), FIRST()))
```

Manufacturer Rank

```
RANK(SUM([annual_sales_units]))
```

Rated Vehicles (so satisfaction ignores nulls explicitly)

```
COUNT([customer_rating])
```

Top N Manufacturers (parameter, integer, default 10, range 5 to 25)

```
[Manufacturer Rank] <= [Top N]
```

Set the Top N parameter control on the dashboard and use the calculated field as a filter on the manufacturer chart.

8. Visualizations (9 Worksheets)

#	Worksheet title	Chart type	Build
1	EV Market Performance at a Glance	KPI cards	One sheet per KPI: SUM(annual_sales_units), AVG(customer_rating), AVG(range_km), AVG(battery_capacity_kwh), COUNTD(manufacturer), COUNTD(model_name). Text mark, large font.

#	Worksheet title	Chart type	Build
2	EV Category Performance and Distribution	Horizontal bar	category on Rows, SUM(annual_sales_units) on Columns, sorted descending, label with Market Share %
3	Market Share by Manufacturer	Horizontal bar	manufacturer on Rows, Market Share % on Columns, Top N filter, colour by AVG(customer_rating)
4	Customer Rating Records by EV Category	Heatmap	category on Rows, Rating Band on Columns, COUNT(vehicle_id) as colour and label
5	Annual EV Sales Trend by Model Year	Line chart	model_year on Columns, SUM(annual_sales_units) on Rows, add category as colour for the drill-down view
6	Battery Technology Market Share	Donut	Pie by battery_type with SUM(annual_sales_units); dual-axis with a white circle to make the donut
7	EV Distribution by Manufacturing Country	Pie chart (or filled map)	manufacturing_country as colour, COUNT(vehicle_id) as angle; filled map is an option
8	Driving Range by Category	Horizontal bar	category on Rows, AVG(range_km) on Columns, with an average reference line
9	Battery Adoption by Category	Heatmap	category on Rows, battery_type on Columns, COUNT(vehicle_id) as colour

[Insert screenshot of each worksheet here]

9. Dashboard Design

Create one dashboard sized Automatic (or Fixed 1400 x 900) with horizontal and vertical containers.

1. **Top row:** title banner and the six KPI cards.

2. **Second row:** Category Performance (left), Battery Technology donut (centre), Manufacturing Country pie (right).
3. **Third row:** Market Share by Manufacturer (left), Annual Sales Trend line (right).
4. **Bottom row:** Rating heatmap, Range by Category and Battery Adoption heatmap.
5. **Right panel:** filters for Manufacturer, Category, Manufacturing Country, Battery Type, Market Region and Model Year, applied to all worksheets (Apply to Worksheets > All Using This Data Source).
6. **Interactivity:** set the Category bar and the Battery donut as Use as Filter, and add a Highlight action on Manufacturing Country.

Colour: one accent colour for battery types (green family), a diverging palette for ratings, and neutral grey for supporting text.

[Insert screenshot of the final dashboard here]

10. Data Story

Build a Tableau Story (Story > New Story) with five scenes, each pairing one worksheet with a caption.

Scene	Caption focus	Worksheet
1	Market overview: 772.6 million units across 200 manufacturers	KPI cards
2	Which categories lead and which are growing	Category bar and sales trend
3	Who leads the market and how satisfied customers are	Manufacturer share and rating heatmap
4	Battery technology mix and how chemistries differ	Donut and battery heatmap
5	Where EVs are built and the outlook	Manufacturing country chart

11. Test Cases and Scenario Mapping

Each test case lists the expected value, computed independently from the CSV, so you can check Tableau against it.

ID	Scenario	Test	Worksheet	Expected result	Pass condition
TC1	1, 2, 3	Dataset loads fully	Data source tab	100,000 rows, 37 columns	Row count matches
TC2	1	Total annual sales	KPI card	772,608,124 units	Equal to the SUM of annual_sales_units
TC3	1	Average customer rating	KPI card	4.08	Rounded to 2 decimals
TC4	1, 3	Average range	KPI card	324.31 km	Rounded to 2 decimals
TC5	1	Top selling category	Category bar	Two-Wheeler Scooter, 85.0M units (11.0%)	Ranked first
TC6	1	Next four categories	Category bar	Sedan 78.3M, Compact SUV 76.1M, Hatchback 76.1M, Three-Wheeler 67.1M	Same order
TC7	1	Lowest selling categories	Category bar	Electric Bus 5.3M, Pickup Truck 6.5M, Electric Truck 6.6M	Bottom three
TC8	2	Leading manufacturer	Manufacturer bar	SurgeWorks, 4.88M units (0.63% share)	Ranked first
TC9	2	Top 10 combined share	Manufacturer bar	6.04%	Sum of top 10 shares
TC10	2, 3	Battery share by sales	Donut	NMC 33.5%, LFP 32.2%, NCA 17.3%, Sodium-Ion 9.8%, Solid-State 7.2%	Slices match
TC11	2	Top manufacturing country by sales	Country chart	China, 78.9M units	Ranked first
TC12	1, 3	Longest-range category	Range bar	Luxury Sedan, 549.2 km	Ranked first
TC13	3	Rating distribution	Rating heatmap	22,393 vehicles above 4.5 and 40,361 between 3 and 4	Counts match
TC14	1	Fastest-growing segment 2018 to 2026	Trend line	Luxury Sedan, +29.7%	Highest Sales Growth %
TC15	1, 2, 3	Filter interaction	Dashboard	Selecting NMC changes every linked sheet	All sheets update, no error

ID	Scenario	Test	Worksheet	Expected result	Pass condition
TC16	2	Top N parameter	Manufacturer bar	Setting 5 shows 5 bars	Bar count equals parameter
TC17	All	Performance with all filters applied	Dashboard	Renders without lag	Interaction under about 3 seconds

12. Results and Insights

Market size and satisfaction. The dataset records 772.6 million annual units across 17,919 model names from 200 manufacturers. Customers rate EVs 4.08 out of 5 on average, and 22.8% of rated vehicles score above 4.5.

Categories. Two-wheeler scooters (11.0%), sedans (10.1%), compact SUVs (9.9%) and hatchbacks (9.9%) contribute the most sales, followed by three-wheelers (8.7%). Affordable urban vehicles lead the market. Electric buses, pickup trucks and electric trucks together make up under 2.4% of sales.

Growth. Comparing model year 2026 with 2018, Luxury Sedans (+29.7%), Electric Buses (+17.2%), Sports Cars (+12.3%), Electric Trucks (+8.2%) and Hatchbacks (+7.2%) grew. Full-size SUVs (-14.9%), Micro City Cars (-12.4%) and Vans / MPVs (-8.5%) declined. Total sales are flat, between 82.6M and 88.1M per year, with a peak in 2021.

Manufacturers. The market is highly fragmented: the top manufacturer holds 0.63% share and the top 10 together hold 6.04%. SurgeWorks, CurrentMobility and ElektraMobility lead on sales. ElektraDynamics (4.17), ChargeWorks (4.15) and WattMotors (4.14) lead on rating among manufacturers with at least 300 records. CurrentMobility pairs high sales with a 4.14 rating.

Battery technology. NMC (33.5% of sales) and LFP (32.2%) dominate. Range is almost identical across chemistries (about 321 to 325 km), but durability differs sharply:

Battery	Share of sales	Avg cycle life	5-year degradation
Solid-State	7.2%	2,999 cycles	7.4%
LFP	32.2%	2,498 cycles	10.1%
Sodium-Ion	9.8%	1,797 cycles	14.0%
NMC	33.5%	1,799 cycles	14.0%

Battery	Share of sales	Avg cycle life	5-year degradation
NCA	17.3%	1,598 cycles	14.0%

Solid-State lasts the longest and degrades the least but has the smallest share, which points to an adoption opportunity. LFP offers the best durability among mass-market chemistries.

Driving range. Luxury Sedans (549 km), Luxury SUVs (525 km) and Full-size SUVs (475 km) offer the longest range; scooters average 105 km and three-wheelers 130 km, consistent with their urban role.

Geography. Manufacturing is spread evenly across all 10 countries: China (78.9M units), Japan (78.3M), Vietnam (78.0M), UK (77.6M) and USA (77.4M) lead, and the lowest, France, still reaches 75.6M. By record count Vietnam is first with 10,172 vehicles. Sales by market region are also even, from 108.3M in Africa to 112.5M in Europe.

Scenario answers. Emily gets category, growth and satisfaction views for portfolio planning. David gets share, battery and country views to position new entrants. Dr. Williams gets the battery lifespan and degradation comparison and the range and manufacturing views.

13. Recommendations

- Keep affordable urban vehicles (scooters, sedans, compact SUVs, hatchbacks) as the volume core of the portfolio.
- Invest in growing premium and commercial segments: Luxury Sedans, Electric Buses, Sports Cars and Electric Trucks.
- Review Full-size SUV and Micro City Car lines, which lost share over the period.
- Because no manufacturer exceeds 0.7% share, new entrants can compete on rating and product quality rather than scale.
- Promote LFP for mass-market durability and treat Solid-State as the premium growth technology.
- Spread production across several countries, as the existing market already does, to lower supply-chain risk.

14. Limitations

- The file is a 100,000-row test extract; totals will differ from the full 200,000-row dataset.
- Each vehicle appears once with its own annual sales, so the model-year trend compares different vehicle models, not the same model over time. Treat growth figures as indicative.
- Category, country, region and battery values are very evenly distributed, so differences between most manufacturers and countries are small.

15. Publishing to Tableau Public

1. Check that the dashboard and story work and delete unused sheets.
2. In Tableau Desktop choose File > Save to Tableau Public and sign in.
3. Name the workbook and click Save; the browser opens the published workbook.
4. Copy the share link or the embed code, and set the workbook to public or hidden as needed.

16. Embedding in a Flask Web Application

The published dashboard can be embedded in a Flask page. Technologies used: HTML for structure, CSS for styling, Tableau Public for hosting and Visual Studio Code for development.

app.py

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route("/")
def home():
    return render_template(
        "index.html",

tableau_url="https://public.tableau.com/views/YOUR_WORKBOOK/YOUR_DASHBOARD?:s
howVizHome=no&:embed=true",
    )

if __name__ == "__main__":
    app.run(debug=True)
```

templates/index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>EV Market Performance and Insights</title>
  <style>
    body { font-family: Arial, sans-serif; margin: 0; background: #f4f7f6; }
    header { background: #0b6e4f; color: #fff; padding: 16px 24px; }
    iframe { width: 100%; height: 900px; border: 0; }
  </style>
</head>
<body>
  <header><h1>Electric Vehicle Market Performance and Insights</h1></header>
  <iframe src="{{ tableau_url }}"></iframe>
</body>
</html>
```

Replace YOUR_WORKBOOK and YOUR_DASHBOARD with your Tableau Public names, then run python app.py and open http://127.0.0.1:5000.

[Insert screenshot of the Flask page here]

17. Conclusion

The EV market in this dataset is large, fragmented and satisfied: 772.6 million units, no manufacturer above 0.63% share and an average rating of 4.08 out of 5. Scooters, sedans, compact SUVs and hatchbacks drive volume, premium and commercial segments are growing, and NMC and LFP batteries lead adoption while Solid-State leads on durability. The dashboard and story give all three stakeholders an interactive way to reach these conclusions and support product, market and policy decisions.